

Question Paper

Exam Date & Time: 21-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
INTRODUCTION TO COMMUNICATION SYSTEMS [ECE_4313]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Give any three examples of industrial domains where 5G-enabled IIoT can be applied effectively. (5)
Why is low latency important for IIoT applications like robotics and process automation in 5G networks?
A)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Illustrate how attenuation and fading can affect communication in a hilly terrain. (3)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- C) Interpret the importance of bandwidth in determining the quality of video streaming services. (2)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) Differentiate between semiconductor Light Emitting Diode (LED) and laser diode as optical sources in terms of power and application. (5)
A)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Use the concept of a link power budget to determine feasibility for a 10 km optical fiber link. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Apply the concept of PIN vs. Avalanche Photodiode (APD) detectors in designing high-speed optical communication links. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Compare the functions of satellite subsystems (power vs. communication vs. control) in practical use. (5)
A)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Apply satellite applications to explain how GPS supports military navigation. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Compare virtual reality (VR) and augmented reality (AR) in terms of user experience and application. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) Illustrate the process of handoff in cellular communication and analyze its necessity when a mobile user transitions between two adjacent cells during an ongoing call. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) Apply the concept of frequency reuse to demonstrate how it improves spectrum efficiency in dense urban networks. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Compare macro cells and micro cells by interpreting their effectiveness in rural and urban environments. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Apply the principle of pulse radar to show how range of a target is measured. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Demonstrate how ZigBee mesh networks are used in smart home applications (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Compare WiMAX with WLAN in terms of coverage and speed. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		ECE_4313:INTRODUCTION TO COMMUNICATION SYSTEMS	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		ECE_4313:INTRODUCTION TO COMMUNICATION SYSTEMS	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	4	12(24%)
3	Application	7	27(54%)
4	Analysis	4	11(22%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)